



U.S. Department
of Transportation
**Federal Aviation
Administration**

Advisory Circular

Subject: Guidance for Parts Manufacturer
Approval of Turbine Engine and Auxiliary
Power Unit Parts under Test and
Computation

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1. Purpose.

a. This advisory circular (AC) provides guidance for developing substantiation data to support the design approval of critical and complex turbine engine and auxiliary power unit (APU) parts produced under parts manufacturer approval (PMA). This guidance is for the comparative test and analysis method used to show compliance to the airworthiness requirements under test and computation, per § 21.303 of Title 14 of the Code of Federal Regulations (14 CFR). This method supports showing the engine or APU still complies with 14 CFR part 33 and Technical Standard Order (TSO) C77.

b. Additionally, this AC provides information on developing a continued operational safety (COS) plan and part numbering for turbine engine parts.

2. Applicability.

a. This AC provides guidance to applicants requesting PMA for critical and complex turbine engine or APU parts. For the purpose of this AC, the term “engine” represents a turbine engine and an APU, and the term “part” represents a turbine engine part and an APU part.

b. This guidance is neither mandatory nor regulatory in nature and does not constitute a regulation. It describes acceptable means, but not the only means, for demonstrating compliance with the applicable regulations. The FAA will consider other methods of demonstrating compliance that an applicant may elect to present. Terms such as “should,” “shall,” “may,” and “must” are used only in the sense of ensuring applicability of this particular method of compliance when the acceptable method of compliance in this document is used. While these guidelines are not mandatory, they are derived from extensive FAA and industry experience in determining compliance with the applicable regulations. On the other hand, if we become aware of circumstances that convince us that following this AC would not result in compliance with the applicable regulations, we will not be bound by the terms of this AC, and we may require additional substantiation as the basis for finding compliance.

c. This document does not change, create any additional, authorize changes in, or permit deviations from, existing regulatory requirements.

3. Related References. Please check the FAA's website at http://www.faa.gov/regulations_policies/ for the latest revision of the following documents.

- a. AC 21-9A, Manufacturers Reporting Failures, Malfunctions, or Defects; May 26, 1982.
- b. AC 21-1B, Production Certificates; May 10, 1976.
- c. AC 33.75-1A, Guidance Material for 14 CFR 33.75, Safety Analysis; September 26, 2007.
- d. AC 39-8, Continued Airworthiness Assessments of Powerplant and Auxiliary Power Unit Installations of Transport Category Airplanes; September 8, 2003.
- e. Order 8110.37D, Designated Engineering Representative (DER) Handbook; August 10, 2006.
- f. Order 8110.42C, Parts Manufacturer Approval Procedures; June 23, 2008.
- g. Order 8110.54, Instructions for Continued Airworthiness Responsibilities, Requirements, and Contents; July 1, 2005.
- h. Order 8120.2F, Production Approval and Certificate Management Procedures; January 30, 2009.
- i. Order 8150.1B, Technical Standard Order Program; May 12, 2002.

4. Background.

a. PMA applicants can show compliance to the airworthiness requirements of the product, i.e., engine certification basis, on which their proposed PMA part is to be installed under test and computation through comparative or general test and analysis, or a combination of both. Applicants requesting PMA for engine parts typically propose comparative test and analysis as the basis for their design approval.

b. When applicants elect to use this comparative test and analysis method to comply with part 33 or TSO C77, the applicant should demonstrate that the functional design of their proposed PMA part is at least equal to that of the original type design part. Applicants frequently use reverse engineering of the type design part to support this approval basis. Although reverse engineering may be adequate to duplicate and substantiate the functional design of a simple engine part, complex or critical engine part designs generally need more rigorous analyses and tests to show equivalency between the type design and the proposed PMA part design. Thus, the test and substantiation plan applicants submit as part of their data package to show compliance

should reflect the criticality or complexity of their proposed PMA part design to ensure the part meets the airworthiness requirements of part 33 or the standards of TSO C77.

c. The FAA is providing the following guidance to assist applicants in developing the substantiation for the design approval of their complex or critical PMA engine parts when using the comparative test and analysis method.

5. Part Criticality. The data supplied by applicants to substantiate that the part meets the requirements of part 33 or TSO C77 varies depending upon the criticality or complexity of the part. This AC uses part categorization as guidance to applicants for determining whether or not a part is critical or complex.

a. Methods for categorizing parts. This AC categorizes parts based on their most severe potential failure effect using various methods for assessing malfunctions and failure modes. Appendix 1 of this AC provides a failure modes and effects assessment suitable for engines. AC 33.75-1A provides descriptions of other acceptable analytical techniques.

b. Part Categories. The categories that this AC addresses are consistent with the previously established categories used in certificate management oversight, as defined by Order 8120.2. These categories cover potential engine failure effects. These categories are also consistent with AC 39-8, which identifies the relative risks of parts and components that could hazard an aircraft if they fail or malfunction, and are based on historic service data and safety assessments. The part categories are defined as follows:

(1) Category 1. A product, i.e. engine, or part(s) thereof, whose failure could prevent continued safe flight and landing; resulting consequences could reduce safety margins, degrade performance, or cause loss of capability to conduct certain flight operations. A Category 1 part, for the purpose of this AC, meets the definition of a critical part defined in Order 8110.42.

(a) We recommend applicants who are considering applying for a PMA of a Category 1 engine part meet with the project aircraft certification office (PACO) to discuss a test and substantiation plan. Early coordination of these projects will help the applicant identify the necessary technical considerations and airworthiness requirements for that part.

(2) Category 2. An engine or part(s) thereof whose failure would not prevent continued safe flight and landing; resulting consequences may reduce the capability of the aircraft or the ability of the crew to cope with adverse operating conditions or subsequent failures. For the purpose of this AC, a Category 2 engine part is typically a complex part that may affect a critical part.

(a) Together with industry, we developed sample templates as aids to identify technical elements and regulatory requirements generally considered when developing test and substantiation plans. These templates are not all inclusive; rather, they are for a limited number of typical Category 2 engine parts. They are included in Appendix 2 of this AC.

(b) For Category 2 engine parts, we recommend applicants discuss any questions regarding the templates with the PACO. If a template is not available for a proposed PMA part, we suggest applicants consider developing a new template to identify technical considerations and regulatory requirements. We recommend applicants stay close to the template format provided to do this. We also recommend applicants review the new template with the PACO early in their project to determine if any additional data will be needed.

(3) Category 3. An engine or part(s) thereof whose failure would have no effect on continued safe flight and landing of the aircraft. The only consequence would be partial or complete loss of engine thrust or power (and associated engine services). For single engine applications, consider changing part categorization to category 1 or 2 if complete loss of thrust could prevent continued safe flight and landing or reduce the ability of the crew to cope with adverse operating conditions or subsequent failures.

c. Tables of Potential Failure Effects. The following tables provide potential failure effects for categories 1 and 2. We also listed some examples of parts whose malfunction or failure could result in one or more of the listed potential failure effects.

Table 1. Category 1

Potential Failure Effects	Part Examples
(1) Non-containment of high-energy debris. (2) Concentration of toxic products in the engine bleed air intended for the cabin, and sufficient to incapacitate crew or passengers. (3) Significant thrust in the opposite direction to that commanded by the pilot. (4) Uncontrolled fire. (5) Failure of the engine mount system leading to inadvertent engine separation. (6) Release of the propeller by the engine, if applicable. (7) Complete inability to shut the engine down.	Life-limited parts. Main engine mounts, with no redundant load carrying features. High pressure vessels (for example, casings subject to compressor discharge pressure and combustor pressure). Containment structures. Fan blades. Fuel system shut-off. Primary structures (for example, structures that provide support and rigidity of the main engine backbone and for attachment of engine to airframe). Thrust reverser control component if thrust reverser control component is part of the engine type certificate (TC).

Table 2. Category 2

Potential Failure Effects	Part Examples
(1) Controlled fires (that is, those brought under control by shutting down the engine or by onboard extinguishing systems). (2) Case burn-through where it can be shown there is no propagation to hazardous engine effects. (3) Release of low-energy parts where it can be shown there is no propagation to hazardous engine effects. (4) Vibration levels that result in crew discomfort. (5) Concentration of toxic products in the engine bleed air for the cabin sufficient to degrade crew performance. (6) Thrust in the opposite direction to that commanded by the pilot, below the level defined as hazardous. (7) Loss of integrity of the load path of the engine supporting system without actual engine separation. (8) Generation of thrust greater than maximum rated thrust. (9) Significant uncontrollable thrust oscillation. (10) Loss of protection such as loss of overspeed protection or loss of containment case capability. (11) Effect or influence on a Category 1 part.	Rotating parts that are not life-limited (for example, compressor and turbine airfoils). Accessory gearbox and internal components. Engine bearings. Spinners. Main engine mounts with redundant load carrying features. Static gas path parts (for example, vanes and seals). Control system actuators. Combustion liners. Fuel nozzles.

6. Comparative Test and Analysis Method. When using this method to show compliance with part 33 or TSO C77, applicants need to show that the functional design of their proposed PMA part is at least equal to that of the type design part. This showing demonstrates that the original findings of compliance for the engine remain valid. The certification basis of the engine identifies the requirements applicable to the proposed PMA part. For critical and complex parts, the applicant should determine which of the original findings could be affected by their proposed PMA part. To do this, applicants must understand how their part functions in the engine operating environment and the associated airworthiness or TSO requirements. Applicants then develop their substantiation data related to these requirements.

a. Functional design. For the purpose of this AC, functional design is a measure of how well the proposed PMA part adheres to the form, fit, and function of the type design part.

(1) Form is the shape, size, dimensions, and other physical measurable parameters that uniquely characterize a part.

(2) Fit is the ability of a part to interface or interconnect, physically and functionally, at the common boundaries with another part or system of the engine.

(3) Function is the action(s) the part is designed to perform in the engine.

(4) Examples of turbine blade functional design include:

- An airfoil design to extract a desired amount of energy at a prescribed speed.
- A blade platform section to provide proper gas path sealing.
- A root section for blade retention that interfaces with the disk.
- Cooling, materials, and other design features that demonstrate the blade is:
 - Durable (resistant to oxidation, corrosion, erosion, creep, fatigue, fracture).
 - Free of destructive vibratory response.
 - Reliable.

b. Substantiation data. Examples of this data include:

- Data based on reverse engineering and formal tests and analyses results;
- Information from engineering handbooks;
- Information based on operational, service, and maintenance experience;
- Reliability data; and
- Other documented information directly applicable to the airworthiness of the proposed PMA part.

Testing must be designed to evaluate the proposed PMA part side-by-side with the type design part to the extent required by those requirements identified above in paragraph 6.

7. Reverse Engineering Considerations. Extracting the functional design of a critical or complex part through reverse engineering requires rigorous testing and analyses. This rigorous evaluation (test and analysis) is necessary to determine the dimensional, material, and processing characteristics needed to develop the proposed PMA part design. This part design consists of drawings and specifications, including a list of drawings and specifications needed to define the configuration and design features of the part. It also includes information on materials, dimensions, and processes necessary to define the structural strength of the part and airworthiness limitations, if required.

a. Reverse engineering a critical or complex part design is far more extensive and difficult than creating a computer-aided-drawing or model of that part. Unintended differences can occur when developing the drawings this way. Applicants should consider a rigorous design and review process to minimize these differences. We recommend applicants use specifications such as American Society of Mechanical Engineers ASME Y14.5M Dimensioning and Tolerancing, or an equivalent specification, for data collection techniques.

(1) Since applicants generally don't know the type design part tolerances before starting a project, determining the measurement method to obtain the required accuracy is a necessary first step. Accuracy is defined as the maximum amount by which the measured result differs from the true value. A general rule of thumb is that measurement accuracy should be at most 10 percent of the expected tolerance. Experience plays a major role in understanding what types of tolerances are used on specific products and features. National standards provided by organizations such as the American Gear Manufacturer's Association (AGMA) or the American Bearing Manufacturer's Association (ABMA), or drawings of similar parts, are useful starting points but should not be used in lieu of actual measurement. Also, a feature's surface finish can provide guidance for required measurement accuracy.

(2) Examples of acceptable measurement accuracies are:

(a) For a roller or ball bearing used in a gas turbine engine, precision measuring equipment capable of measuring to an accuracy of 0.00001 inch or better.

(b) For a precision ground surface used as a land for a carbon seal, optical flats to get the required flatness.

b. When reverse engineering, we recommend taking measurements from multiple type design samples to create the tolerance bands for the drawings and specifications of the proposed PMA part. These tolerance bands tend to be tighter than those of the type design, particularly if based on a limited sample size. A purely statistical method for determining the sample size may result in a sample size which is extensive but not practical. The drawing tolerances determined by reverse engineering for the proposed PMA part should not exceed the measured data (unless substantiated). Applicants should verify that their proposed PMA part can be manufactured to these tighter tolerances. If tighter tolerances cannot be achieved, then the applicant may either:

(1) Analyze more type design parts to expand the reverse-engineered tolerances, change the drawing requirements, and support the manufacturing process; or

(2) Use a more precise manufacturing method (for example, grinding versus milling) to bring the proposed manufactured part within tolerances.

c. A reverse-engineered PMA part is intended to duplicate the form, fit, and function of the type design part it is planned to replace. Therefore, reverse engineering a part or an assembly is not limited to copying the physical type design part. Some aspects of the type design cannot be revealed through physical observation. For example, the outcome of certain manufacturing processes, such as heat treat and shot peening, can only be evaluated by destructive inspection techniques. Therefore, these types of manufacturing processes must be carefully controlled to ensure the produced PMA part is at least equal to the type design part throughout manufacture. Table 3 provides examples of manufacturing processes that may require control. For PMA parts that have controlled manufacturing processes, we recommend applicants provide their plans for validating these processes at the time of their PMA application. We refer to these plans as source substantiation plans.

Table 3. Typical Controlled Manufacturing Processes

1. Abrasive finishing, for example, tumbling and machining
2. Anodizing and hardcoat
3. Brazing
4. Case hardening (case carburizing)
5. Casting processes
6. Chemical cleaning
7. Cold working, for example, bending, straightening, and hammering
8. Conventional machining (feeds, speeds, and tool wear)
9. Diffusion coating (aluminizing)
10. Dry film lube (application of)
11. Electron beam (EB) weld
12. Electro discharge machining (EDM)
13. Forging processes
14. Grit and abrasive blasting (wet or dry)
15. Heat treatment
16. Hot forming
17. Laser drilling (cutting and marking)
18. Plating
19. Rust inhibiting
20. Shot peening
21. Spot and seam welding
22. Spark grinding
23. Titanium chemical cleaning
24. Tungsten inert gas (TIG) welding
25. Tooling design, for example, dies and grip fixtures
26. Vapor degreasing
27. Electro chemical machining (ECM)
28. Composite material manufacturing processes
29. Media (Mass) finish
30. Plasma spraying
31. Titanium welding
32. Bi-metallic welding
33. Electrochemical marking
34. Laser shock peening
35. Elastomeric material processes, for example, curing and mixing
36. Ultrasonic and eddy current inspection (ECI)
37. Inertia welding
38. Translational friction welding
39. Friction stir welding
40. Hole drilling of high length over diameter (L/D) holes
41. Broaching
42. Material removal in general for rotating parts

d. The part design extracted through reverse engineering data, along with other factors such as service experience, may form the basis on which improvements to features or characteristics are introduced. However, intentional deviation from the type design of critical and complex parts can raise concerns regarding configuration management issues, unintended consequences of the design change, and applicability of the existing instructions for continued airworthiness (ICA). Therefore, applicants should consider introducing a change to their proposed PMA part design only after completing a detailed examination and design evaluation to determine if the change constitutes a major design change per § 21.93. Applicants must clearly identify and justify through engineering evaluation all intentional changes to the proposed PMA part design with the ACO engineer(s) reviewing the design approval.

(1) Configuration management issues can occur because the TC holder can make changes to their part or assembly designs without considering alternate PMA-approved configurations. For instance, if a PMA part adds a disassembly aid feature, such as a tab, and subsequently the TC holder makes a design change to the mating part that creates an interference with the tab on the PMA part, then the PMA part would not be upwardly compatible with the TC holder's design change.

(2) Unintended consequences may also occur when applicants make changes to their proposed PMA part design. For example, if the PMA holder alters their turbine blade design to make the dovetail stronger, this could change the "weak link" in the system from the blade to the disk post. Consequently, a disk post failure of the PMA part could result in the release of at least two blades and a disk post instead of a single blade, which was the type design intent. Thus, applicants must have a thorough understanding of the relationship between their proposed changes and the assembly in which their part is installed.

(3) Applicants must account for any differences between their proposed PMA part and the type design part when assessing the applicability of the existing ICA, including repair processes. When the proposed PMA part can be installed in multiple engine models, we recommend that applicants consider the most limiting operating or installation conditions, or both, such as:

- Installation applications, such as single engine versus multi-engine aircraft.
- Engines approved for Extended Operations (ETOPS).
- Recommended operating limitations, such as
 - Time Between Overhaul,
 - Thrust and power ratings, and
 - Redlines.

e. An outline of a reverse engineering design process for complex parts is provided in Appendix 3.

8. Continued Operational Safety. PMA holders must support the continued operational safety of their designs. We recommend applicants develop a COS plan to do this. To establish an effective COS plan, applicants should have the ability to assess design, manufacturing, and maintenance issues related to the operation of the engine on which their proposed PMA part is installed.

a. A COS plan generally includes a suitable management plan that includes continuous assessment of the part's performance in service relative to the applicant's design assumptions. If a failure, malfunction, or defect in service is identified, the PMA holder must report it as required by § 21.3.

b. A COS plan also includes appropriate methods and resources to identify the cause of a failure, malfunction or in-service defect, develop corrective actions, and implement those actions in a timely manner. Applicants should also validate that the corrective actions restore the engine to an acceptable level of safety.

c. Appendix 4 of this AC provides additional information on how to develop a successful COS plan. If applicants elect to submit a COS plan, they should submit their plan during the design approval phase of the PMA process.

9. Part Numbering. When applicants assign a part number (P/N) to their PMA part, we recommend they use a suffix or prefix of at least 2 letters. The use of a single letter prefix or suffix to the type design part number may result in confusion between the PMA and type design parts. Engine TC holder part numbering practices often employ a single letter prefix or suffix. For example, P/Ns 123456A and 123456B, or A123456 and B123456, designate certain changes to the type design part. Since applicants generally do not have complete knowledge of the type design part number evolution, they should use care when selecting the proposed PMA part number.



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APPENDIX 1. FAILURE MODES AND EFFECTS ASSESSMENT

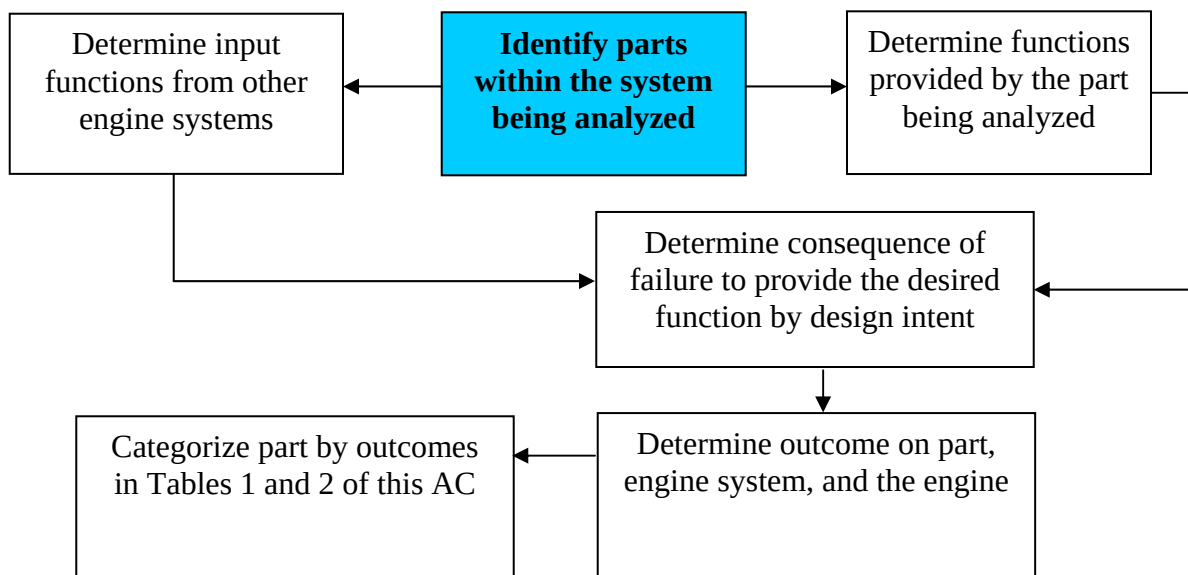
1. A failure modes and effects assessment is a qualitative process, independent of failure rates and probabilities, by which each failure mode of a part in the engine system is analyzed. Some top-level functions typically considered in an engine failure modes and effects assessment are:

- Maintaining structural integrity, including allowed overspeed and overtemperature exceedances;
- Providing thrust or power;
- Operating in inclement weather;
- Providing customer bleed, power extraction, or both; and
- Meeting fuel consumption, exhaust gas temperature, vibration, emission or noise limits.

2. Each system and subsystem of the engine is broken down into its basic functions using a functional block diagram consistent with the Air Transport Association policy for identification and definition of systems.

3. The functional block diagram defines each system and subsystem, and all their functions, in the turbine engine. The experienced safety engineer performing the analysis determines the part-to-part and part-to-system influences in both directions (input and output). The process flow is shown in figure A1.1 below:

Figure A1.1. Process Flow Diagram



4. The part categorization process is built around the fundamental understanding of the part function and its potential effects on physically or functionally mating parts or both. The fundamental premise in the categorization process, and in the physical operation and function of the turbine engine, is system interactions.

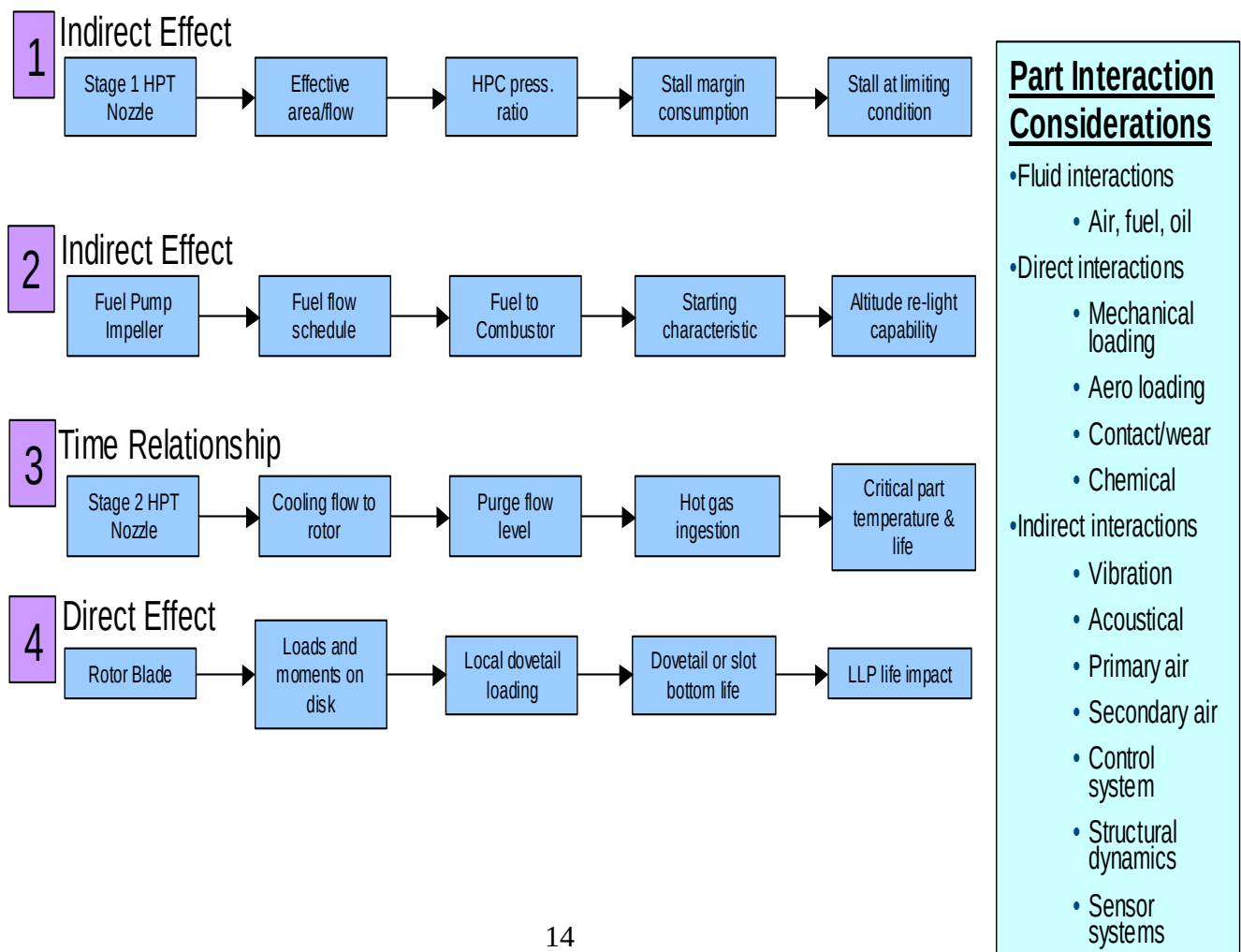
a. System interactions are influences a part, or a set of parts, can have on the turbine engine, propulsion system, or aircraft through form, fit, or function. These influences may extend beyond the component being classified, may be direct or indirect, and may develop immediately or over time. Characteristics of these influences include:

(1) Direct influences, which are form and fit. These influences are based on physical contact or interface clearances between adjacent parts.

(2) Indirect influences, which are functional in nature. These influences are not based on physical contact, but may be aerodynamic, thermal, or vibratory.

5. The interactions where the consequence of failure is the furthest from the cause are the most difficult to identify. Many fundamental relationships in part interactions and subsequent system effects exist. Figure A1.2 below provides four examples.

Figure A1.2. Part Interaction Considerations



APPENDIX 2. TEMPLATES

1. This appendix provides 17 templates, listed in Table A2-1 below. The FAA selected them based on what we found applicants were typically submitting for complex turbine engine part PMA approval. These templates will aid applicants in identifying the technical elements and regulatory requirements they should consider when developing their test and substantiation plans. Applicants must ensure that any additional technical criteria or regulatory requirements are met for their specific proposed PMA part. Each template is subdivided into the following sections:
 - a. Section 1 - Part functional capability, which is the action the part is designed to perform in the engine.
 - b. Section 2 - Part shape, size, dimensions, and other physical, measurable parameters; interconnectivity with an engine integral part or system; or both.
 - c. Section 3 – Part materials and processes – properties.
 - d. Section 4 – Part materials and processes – processing.
 - e. Section 5 - Applicable analyses.
 - f. Section 6 - Applicable testing.
 - g. Section 7 - Applicable regulatory requirements.
2. Sections 1 through 5 are divided into the following five columns:
 - a. Column 1 is the section number.
 - b. Columns 2 and 3 identify part characteristics to be evaluated to support the functional design intent of the type design part.
 - c. Column 4 identifies typical areas applicants should evaluate for potential part and system failure modes.
 - d. Column 5 is used to identify a combination of actions and criteria for risk mitigation and control when a corresponding failure mode is identified in column 4.
3. Section 7 of the templates identifies the part 33 (Amendments 1-20 inclusive) airworthiness requirements applicable to turbine engines. However, not all engine certification bases include the same requirements.
4. If a template is not available for a particular part, the applicant can create or modify another as necessary. For example, although templates specific to APU parts are not provided, the

applicant may use the templates for similar APU parts and revise the regulatory section to reflect the requirements of TSO C77.

Table A2-1. Available Templates

Template Number	Part Name
1	O-Rings
2	Embedded elastomer seals
3	Combustor
4	Low pressure turbine blade
5	Turbine vane cooled
6	Turbine vane un-cooled
7	Shaft
8	Roller bearing
9	Ball bearing
10	Gear
11	Compressor vane
12	Compressor blade
13	Static air seal
14	Variable stator vane lever arms
15	Bushings
16	Turbine blade cooled
17	Fuel filter